International Scientific Physics Olympiad

T2. Muon anomalous magnetic moment

Answer sheet

Part A. Dynamics in magnetic field (4.5 points)

A1 (Formula) $\Delta v =$ (Value) $\Delta v =$ **1.0**

A2 (Formula) $E_k =$ (Value) $E_k =$ **0.5**

A3 (Formula) $R =$ (Value) $R =$ **1.0**

(Formula) $T =$ (Value) $T =$

A4 (Formula) $N =$ (Value) $N =$ **0.8**

A5 $d\vec{s}/dt =$ **0.7**

A6 $\omega_z =$ **0.5**

Part B. Thomas precession (3.5 points)

B1 $v_{Cx} =$ $v_{Cy} =$ **0.5**

B2 $v''_{x''} =$ $v''_{y''} =$ **0.7**

B3 **0.6**

B4 $\omega =$ **1.0**

B5 $\omega_{T,z} =$ **0.7**

Part C. Magnetic moment precession (2 points)

C1 **0.7**

$\omega_s =$

C2 **0.5**

$\omega_a =$

C3 **0.8**

(Formula) $a_\mu =$ (Value) $a_\mu =$